Module 08 Lab Reflective Journal

John Grubbs

I liked this assignment because it helped me understand why a model can perform well on training data but still make poor predictions on new data. The graphs made the ideas of underfitting, overfitting, and a good fit easier to understand.

I learned that a degree 1 polynomial underfits because it is too simple to follow the pattern in the data. A degree 15 polynomial overfits because it follows the individual data points too closely and creates a jagged line on the graph. The degree 4 polynomial was the best fit because it followed the overall pattern without memorizing the noise.

The learning curves also helped me understand the bias-variance tradeoff. An underfitting model has low training and validation scores because it is too simple. An overfitting model has a large gap between its high training score and lower validation score. This showed me why models need to generalize instead of simply memorizing the training data.

Luis Adrian

This lab helped me understand how model complexity affects learning. Comparing polynomial models with degrees 1, 4, and 15 showed why a simple model may miss important patterns, while a complex model may fit noise. The sine wave dataset made these concepts easier to visualize and helped me connect underfitting with high bias and overfitting with high variance.

I also learned why training performance alone is not enough to evaluate a model. Learning curves compare training and cross-validation scores to help identify generalization problems. This activity connected to previous regression labs by showing that choosing a model requires balancing complexity and validation performance. In future projects, I would examine both scores and the data split before deciding which model performs best.

Keira Burdine

In this lab, I learned about the bias-variance tradeoff and how model complexity affects machine learning performance. I worked with polynomial regression models using degrees 1,4 and 15 to understand the differences between underfitting, a good fit and overfitting. I also created learning curves to compare training and cross-validation scores.

One of my biggest challenges was correcting syntax errors and indention issues in Google Colab. Some of my code initially produced errors or did not display the graphs correctly. After troubleshooting and making adjustments, I was able to generate the graphs and complete the

lab. This experience helped me better understand the importance of properly structured Python code and how to interpret model performance. Overall, this lab strengthened my understanding of machine learning and improved my confidence in troubleshooting coding errors.